

Department of Computer Science
CS403NOD – Node.js

Lab Assignment – 03

Student Directory API — Working with Route Parameters

Semester: BCA VII

Objective

After completing this task, you should be able to:

- Read a value directly out of the URL to build a dynamic route.
- Return specific data based on that value, instead of one fixed response.
- Use array methods (find, filter) with real request data.
- Handle “not found” cases gracefully with the correct status code.
- Extend a working Node.js server to a new, realistic use case.

Tasks

Task 1 – Project Setup

Continue inside your existing Project folder from today's main lab. Add the following files.

```
NodeJS-Lab/  
└── Lab-03/  
    ├── students-server.js  
    ├── students-output.png  
    └── items-output.png
```

Task 2 – Create The Student Directory Server

Create

students-server.js

Use the code below as your starting point.

```
const http = require('http');  
  
const students = [  
  { id: 1, name: "Aman", course: "BCA" },  
  { id: 2, name: "Nandani", course: "BCA" },  
  { id: 3, name: "Abhishek", course: "BIT" }  
];  
  
const server = http.createServer((req, res) => {  
  res.setHeader('Content-Type', 'application/json');  
  
  if (req.url === '/students') {  
    res.end(JSON.stringify(students));  
  }  
  else if (req.url.startsWith('/students/')) {  
    const id = Number(req.url.split('/')[2]);  
    const student = students.find(s => s.id === id);  
  
    if (student) {  
      res.end(JSON.stringify(student));  
    } else {  
      res.writeHead(404);  
      res.end(JSON.stringify({ error: "Student not found" }));  
    }  
  }  
});
```

```

    }
    else {
      res.writeHead(404);
      res.end(JSON.stringify({ error: "Route not found" }));
    }
  });

  server.listen(3000, () => console.log("Server running on port 3000"));

```

Run the server.

node students-server.js

Test the following in your browser:

- /students
- /students/1
- /students/2
- /students/99 (should show "Student not found")

Take a screenshot showing at least two of these outputs.

Save it as

students-output.png

Task 3 – Build Your Own Directory

Replace the students array with your own array of at least 5 items. Choose any theme — books, movies, products, cricket players, anything you like.

Create matching routes:

- /items — returns the full list
- /items/:id — returns one item by id, or a "not found" message

Test all routes, then take a screenshot.

Save it as

items-output.png

Task 4 – Bonus Challenge

- Add /students/course/BCA that returns only students from that course, using `Array.filter()`.
- Handle a non-numeric id gracefully — e.g. /students/abc should return a clear error message instead of a confusing result.

README.md

Update your existing README.md to also list:

- The new routes you added today and what each one returns
- A one-line note on what `req.url.split()` is doing in your code

GitHub Submission

Use your existing repository — no need to create a new one.

Step 1

Open Terminal inside NodeJS-Lab.

Step 2

Run the following commands.

git add .

git commit -m "Lab 03 - Student Directory API"

git push

Google Classroom Submission

Submit

- GitHub Repository Link
- (Optional) ZIP File

If You Face Any Problem

If you are unable to complete any task, **do not leave it incomplete without explanation.**

Create a section in your **README.md** named:

Problems Faced

Mention:

- Task Number
- Error / Issue
- What you tried to solve it
- Screenshot (if applicable)

Example

Problems Faced

Task No.: Task 2

Issue:

Visiting /students/2 always returns the first student, no matter the id.

Attempted Solution:

Realized req.url.split('/')[2] was actually the id, but I had written [1] by mistake. Fixed the index.

Students who clearly document genuine problems will receive guidance during the next lab session.